



Solution Guide: Restoring Data with Incremental Backups

Incremental Backup Restoration

1. Move into the `~/Documents/epscript/testenvir` directory and list the contents.
 - Run `cd ~/Documents/epscript/testenvir`.
 - Run `ls -l`.
2. Notice the `doctor`, `patient`, and `treatment` directories.

Unset

```
total 12
drwxr-xr-x 2 sysadmin sysadmin 4096 Jul 14 10:14 doctor
drwxr-xr-x 2 sysadmin sysadmin 4096 Jul 14 10:14 patient
drwxr-xr-x 3 sysadmin sysadmin 4096 Jul 14 10:14 treatment
```

3. In your `~/Documents/epscript` directory, create the level 0 backup of the `testenvir` directory, which contains the `doctor`, `patient`, and `treatment` directories.
 - Move back into the `epscript` directory:
 - Run `cd ../`.
 - Create the level 0 backup of the `testenvir` directory by running:
 - Run `tar cvvWf epscript_back_sun.tar --listed-incremental=epscript_backup.snar --level=0 testenvir`.
 - Your very verbose output should look similar to the following:

Unset

```
tar: testenvir: Directory is new
tar: testenvir/doctor: Directory is new
tar: testenvir/patient: Directory is new
tar: testenvir/treatment: Directory is new
tar: testenvir/treatment/backup: Directory is new
```

```
drwxr-xr-x sysadmin/sysadmin 0 2020-07-14 10:14 testenvir/  
drwxr-xr-x sysadmin/sysadmin 0 2020-07-14 10:14 testenvir/doctor/  
drwxr-xr-x sysadmin/sysadmin 0 2020-07-14 10:14 testenvir/patient/  
drwxr-xr-x sysadmin/sysadmin 0 2020-07-14 10:14 testenvir/treatment/
```

4. We can view and verify the contents of the **level 0** backup by using `tar` as follows:
- Run `tar tvvf epscript_back_sun.tar --incremental | less`.
 - Tap the `tab` button on the keyboard to advance the screen one page at a time.
 - Tap the `enter` key to advance one line at a time.
 - Status of the files in the backup should look similar to the following:

Unset

```
drwxr-xr-x sysadmin/sysadmin 29 2020-07-14 10:14 testenvir/  
D doctor  
D patient  
D treatment  
  
drwxr-xr-x sysadmin/sysadmin 325 2020-07-14 10:14 testenvir/doctor/  
Y doctors.1.csv  
Y doctors.10.csv  
Y doctors.11.csv  
  
...  
...  
  
drwxr-xr-x sysadmin/sysadmin 346 2020-07-14 10:14 testenvir/patient/  
Y patients.1.csv  
Y patients.10.csv  
Y patients.11.csv  
  
...  
...  
  
drwxr-xr-x sysadmin/sysadmin 396 2020-07-14 10:14  
testenvir/treatment/  
D backup
```

```
Y treatments.1.csv
Y treatments.10.csv
Y treatments.11.csv
```

```
...
...
```

- What is the status of the files in the backup?
 - **D** indicates directories.
 - **Y** indicates that these files are contained in the `epscript_back_sun.tar` archive.
- 5. Simulate a natural disaster or cyberattack by removing the `patient` directory.
 - From the `~/Documents/epscript` directory:
 - Run `rm -r testenvir/patient/`.
 - Verify that the `patient` directory is removed:
 - Run `ls -l testenvir/`.
 - Your output should look similar to the following. Notice that the `patient` directory is missing:

Unset

```
total 8
drwxr-xr-x 2 sysadmin sysadmin 4096 Jul 14 10:14 doctor
drwxr-xr-x 3 sysadmin sysadmin 4096 Jul 14 10:14 treatment
```

- 6. Restore the missing `patient` directory from the `epscript_back_sun.tar` backup to the `~/Documents/epscript/testenvir/patient/` directory.
 - Make sure you're in the `epscript` directory:
 - Run `cd ~/Documents/epscript`.
 - Restore the missing **patient** directory:
 - Run `tar xvf epscript_back_sun.tar -R -C ~/Documents/epscript/ testenvir/patient/`.
 - Your very verbose output should look similar to the following:

Unset

```
block 5: drwxr-xr-x sysadmin/sysadmin 346 2020-07-14 10:14
testenvir/patient/
```

```
block 341: -rw-r--r-- sysadmin/sysadmin 6329 2020-07-14 10:14
testenvir/patient/patients.1.csv
block 355: -rw-r--r-- sysadmin/sysadmin 6236 2020-07-14 10:14
testenvir/patient/patients.10.csv
block 369: -rw-r--r-- sysadmin/sysadmin 6250 2020-07-14 10:14
testenvir/patient/patients.11.csv
block 383: -rw-r--r-- sysadmin/sysadmin 6311 2020-07-14 10:14
testenvir/patient/patients.12.csv
```

- Verify that the files have been added to the `testenvir/patient` directory successfully.
 - Run `ls -l testenvir/`.
- Your output should look similar to the following:

Unset

```
total 12
drwxr-xr-x 2 sysadmin sysadmin 4096 Jul 14 10:14 doctor
drwxr-xr-x 2 sysadmin sysadmin 4096 Jul 14 10:14 patient
drwxr-xr-x 3 sysadmin sysadmin 4096 Jul 14 10:14 treatment
```

- The missing `patient` directory has been properly restored from the archive.

7. Before we create an incremental backup, we'll create some files in the `patient` directory:

- Make sure you're in the `cd ~/Documents/epscript/testenvir/patient/` directory.
 - Run `cd ~/Documents/epscript/testenvir/patient/`.
- Create a couple of arbitrary files:
 - Run `touch patient.0a.txt patient.0b.txt`.
- Verify that the files have been added:
 - Run `ls -l`.
- Output should look similar to the following. Notice the two new files `patient.0a.txt` and `patient.0b.txt` have been successfully created:

Unset

```
total 284
-rw-r--r-- 1 sysadmin sysadmin      0 Aug 13 12:27 patient.0a.txt
-rw-r--r-- 1 sysadmin sysadmin      0 Aug 13 12:27 patient.0b.txt
```

```
-rw-r--r-- 1 sysadmin sysadmin 6236 Jul 14 10:14 patients.10.csv
-rw-r--r-- 1 sysadmin sysadmin 6250 Jul 14 10:14 patients.11.csv
```

8. Assume it's Monday. Now, we'll create an incremental backup that will include our newly created patient documents as follows:
- Change back to the `~/Documents/epscript` directory:
 - Run `cd ~/Documents/epscript`.
 - Create an incremental backup for Monday as follows:
 - Run `tar cvvWf epscript_back_mon.tar --listed-incremental=epscript_backup.snar testenvir`.
 - List the contents of the `epscript_back_mon.tar` incremental backup and verify that the new files have been archived.
 - Run `tar tvvf epscript_back_mon.tar --incremental | less`.
 - Your output should look similar to the following. Notice that our new patient files `patient.0a.txt` and `patient.0b.txt` have been successfully archived under the `testenvir/patient/` directory:

Unset

```
drwxr-xr-x sysadmin/sysadmin 29 2020-08-13 12:05 testenvir/
D doctor
D patient
D treatment
```

```
drwxr-xr-x sysadmin/sysadmin 325 2020-07-14 10:14 testenvir/doctor/
N doctors.1.csv
N doctors.10.csv
N doctors.11.csv
N doctors.12.csv
```

```
...
...
```

```
drwxr-xr-x sysadmin/sysadmin 378 2020-08-13 12:27 testenvir/patient/
Y patient.0a.txt
Y patient.0b.txt
Y patients.1.csv
```

```

Y patients.10.csv
Y patients.11.csv

...

drwxr-xr-x sysadmin/sysadmin 396 2020-07-14 10:14
testenvir/treatment/
D backup
N treatments.1.csv
N treatments.10.csv
N treatments.11.csv
N treatments.12.csv

...

```

- What is the status of the files in the incremental backup?
 - **D** indicates directories.
 - **N** indicates that the file was present in the directory at the time the archive was made. However, it was not added to the `epscript_back_mon.tar` archive because it had not changed since the last backup.
 - **Y** indicates that the file is contained in the `epscript_back_mon.tar` archive.

Bonus Review Questions

1. What is the difference between a `full` and `incremental` backup?
 - A **full backup** saves every file on a hard drive.
 - An **incremental backup** only saves the data that has changed since the last full backup.
2. If you have a backup schedule of Monday, Tuesday, Wednesday, Thursday, and Friday:
 - On what day would you schedule a level 0 backup to be done?
 - Monday
 - In what order should the backups be applied to restore a system that was completely lost after an attack?
 - Start with Monday, end with Friday.
3. What command do you use to create a level 0 backup of `archive/home/user1`?
 - `tar cvvWf backup.tar --listed-incremental=backup.snar --level 0 archive/home/user1`
4. What command would you use to list the contents of an incremental backup?

- `tar tvvf backup.tar --incremental`
- 5. After listing the contents of an incremental backup, what do the following letters indicate:
 - **x** indicates that the file is contained in the `backup.tar` archive.
 - **n** indicates that the file was present in the directory at the time the archive was made, but was not added to the `backup.tar` archive because it has not changed since the last backup.
 - **D** indicates the file is a directory.